

Program Studi Teknik Otomasi

Jurusan Teknik Elektro

Politeknik Negeri Bali

2025-2026



POLITEKNIK NEGERI BALI

MATA KULIAH KONTROL CERDAS (MKB2436304405)

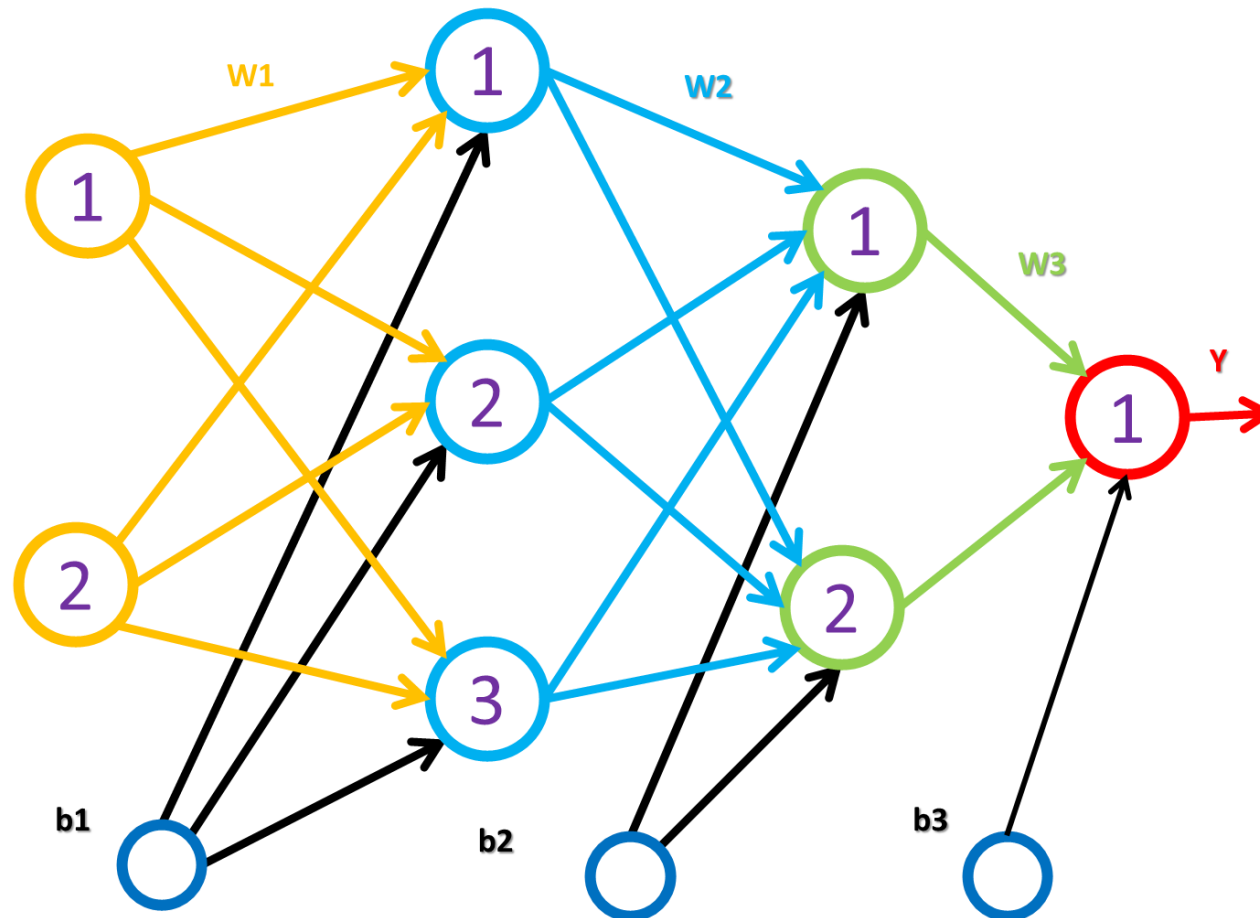
Mengenali Bobot & Bias

Dosen Pengampu:

Dr. I N. K. Wardana

Contoh

- ❑ Buatlah network untuk menirukan fungsi **OR**.
- ❑ Gunakan arsitektur berikut:



Pemrograman MATLAB

```
clc; clear; close all;
```

```
% DATA LOGIKA OR
```

```
X = [0 0 1 1; 0 1 0 1]; % input
```

```
T = [0 1 1 1]; % target
```

```
% BANGUN JARINGAN MLP
```

```
net = feedforwardnet([3 2]);
```

```
% Pengaturan fungsi aktivasi
```

```
net.layers{1}.transferFcn = 'logsig'; % hidden layer 1
```

```
net.layers{2}.transferFcn = 'logsig'; % hidden layer 2
```

```
net.layers{3}.transferFcn = 'logsig'; % output layer
```

```
% Algoritma training
```

```
net.trainFcn = 'trainlm'; % Levenberg-Marquardt
```

```
net.trainParam.epochs = 1000; % jumlah iterasi
```

```
net.trainParam.goal = 1e-5; % target error
```

```
% Pembagian data (semua untuk training, dataset kecil)
```

```
net.divideParam.trainRatio = 1;
```

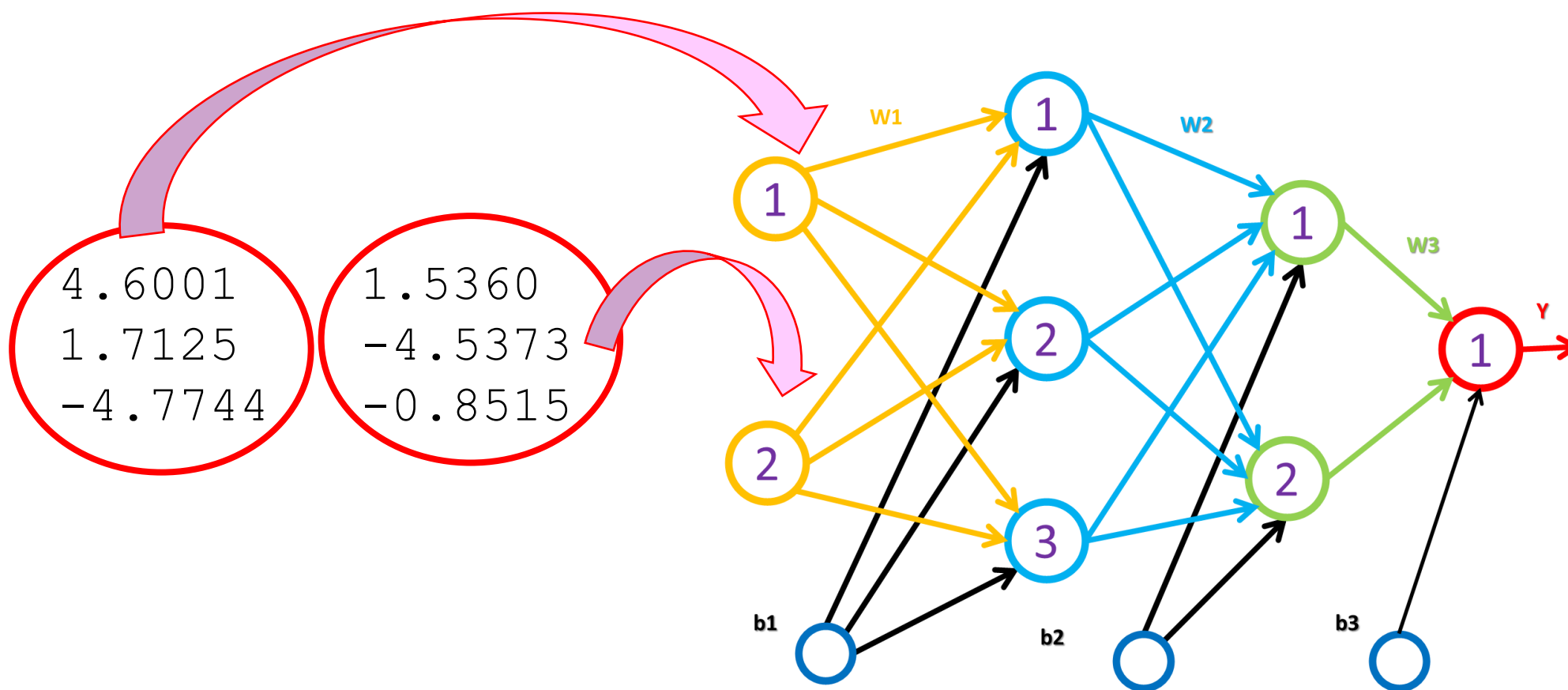
```
net.divideParam.valRatio = 0;
```

```
net.divideParam.testRatio = 0;
```

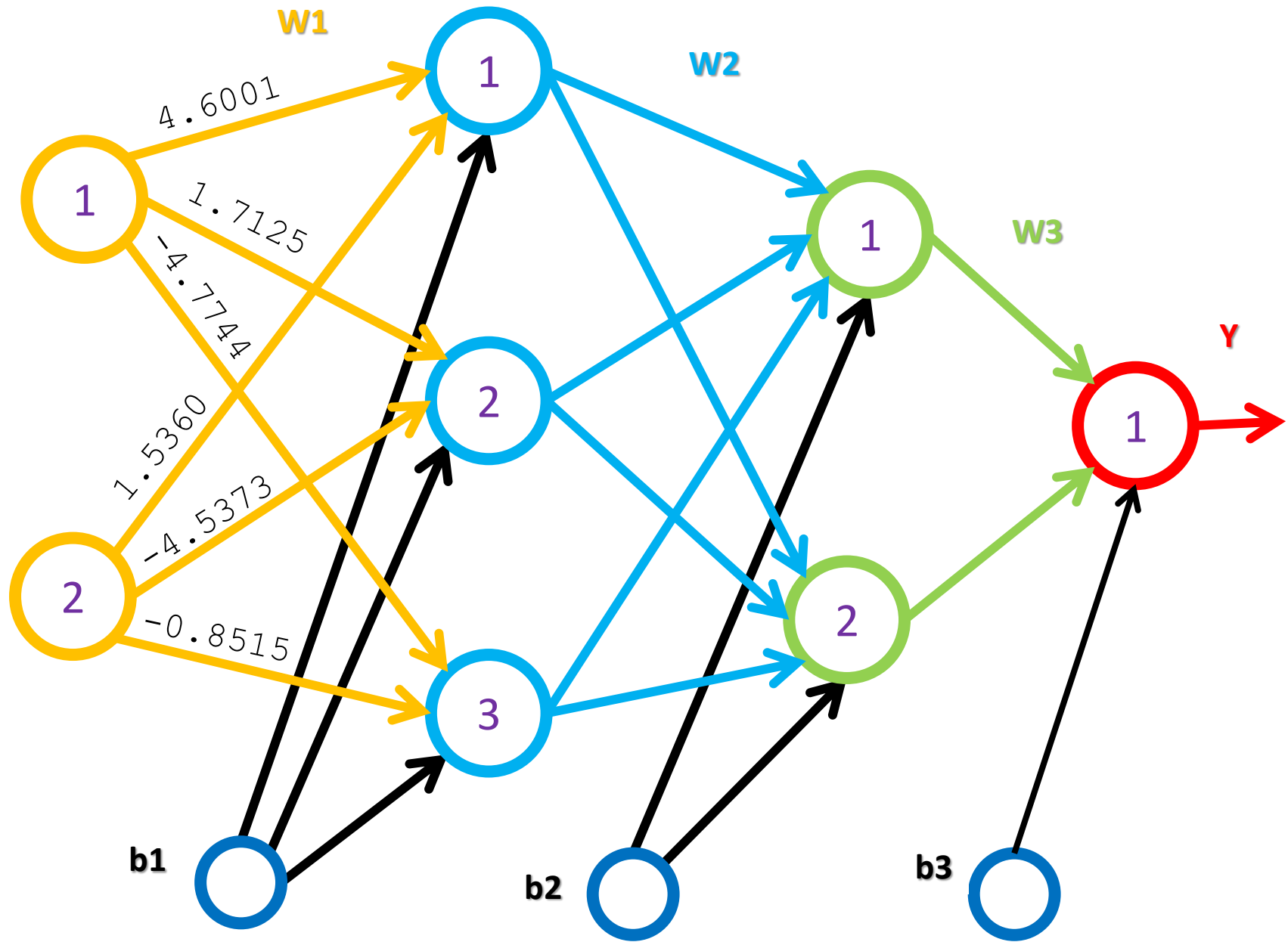
```
% --- Inisialisasi konsisten ---  
rng(3);  
net = configure(net, X, T);  
net = init(net);
```

```

>> % Bobot awal
>> net.IW{1,1}
    
```



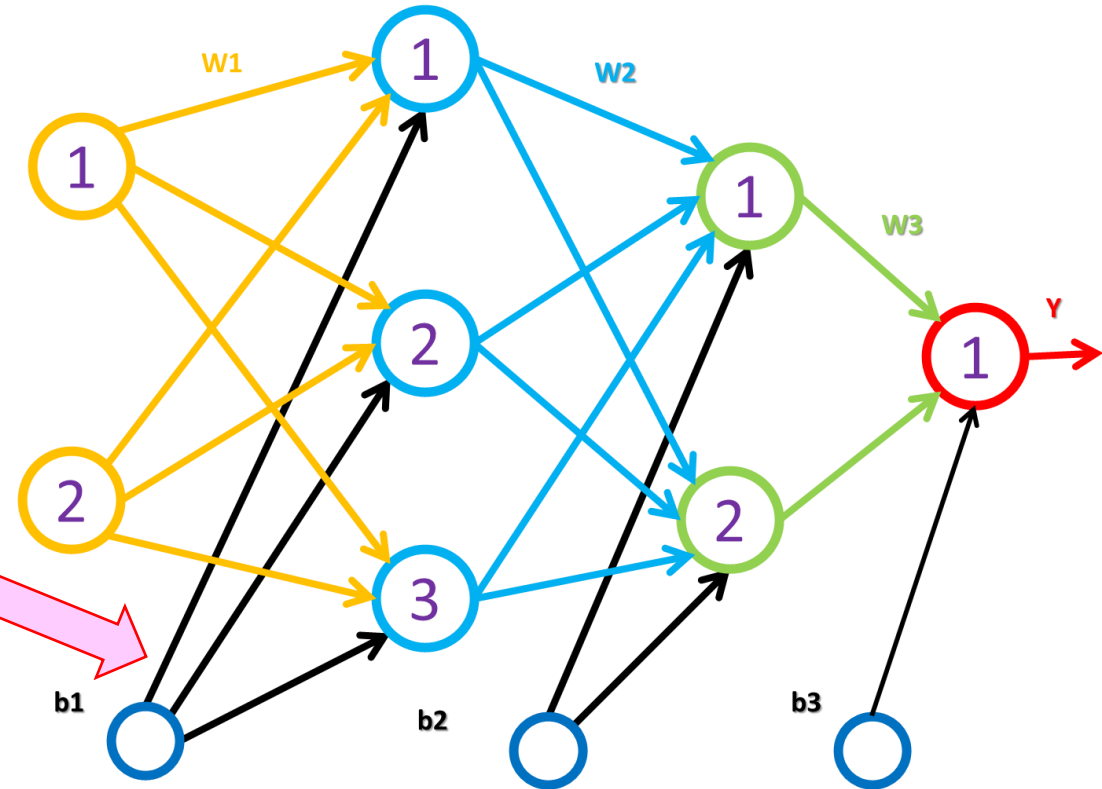
4.6001	1.5360
1.7125	-4.5373
-4.7744	-0.8515



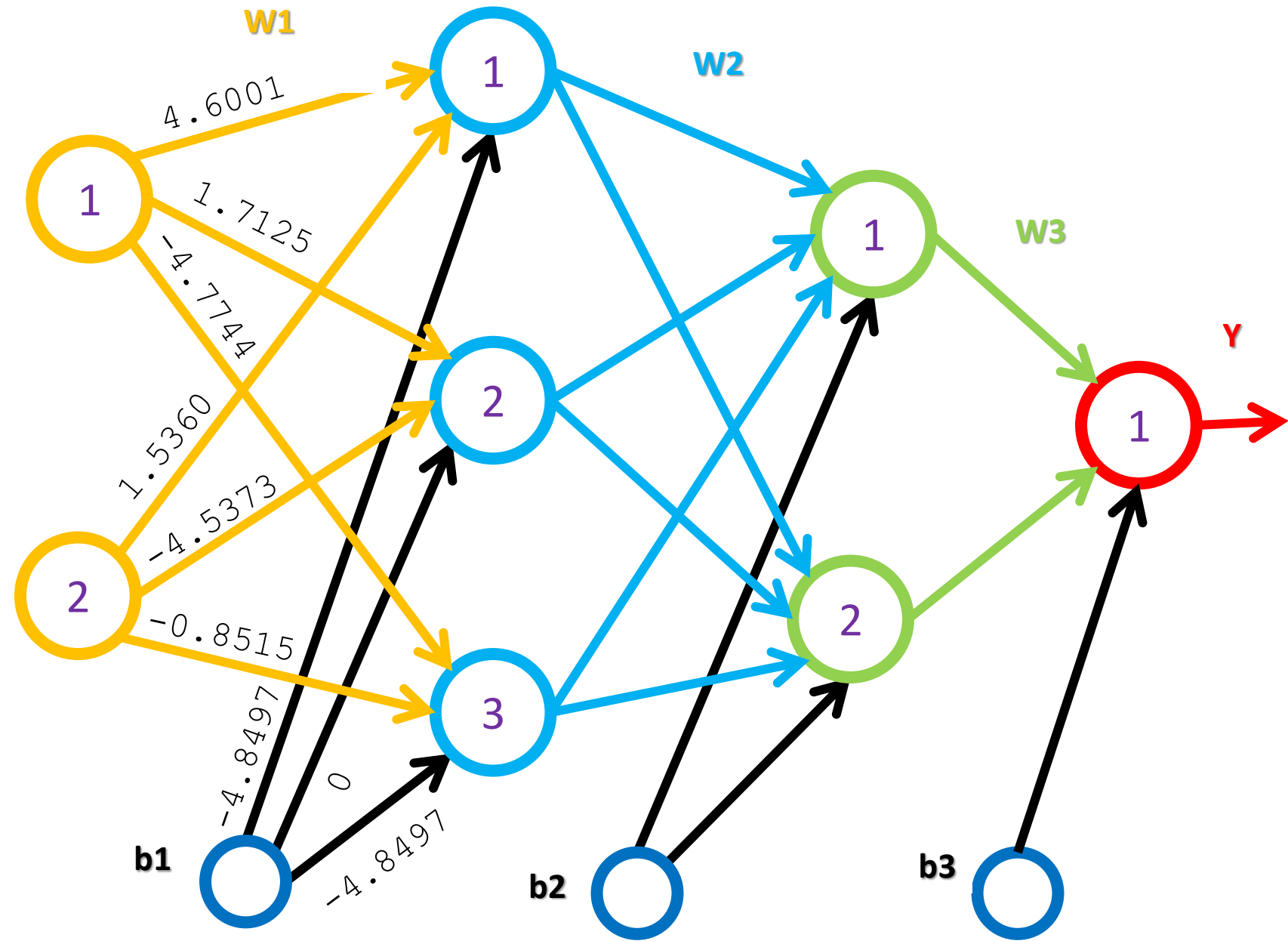
```
>> % Lihat bias di H1
>> net.b{1}
```

ans =

-4.8497
0
-4.8497



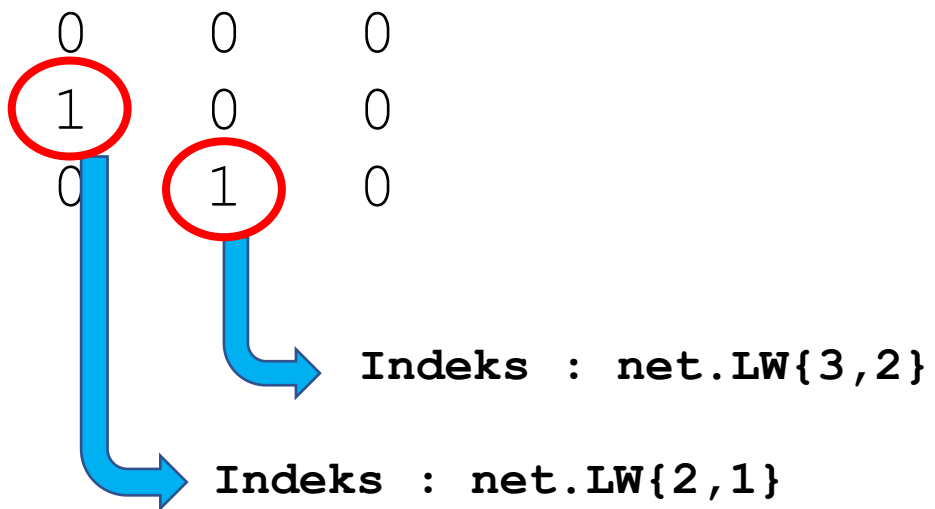
-4.8497
0
-4.8497



```
>> % Cek koneksi layer  
>> net.layerConnect
```

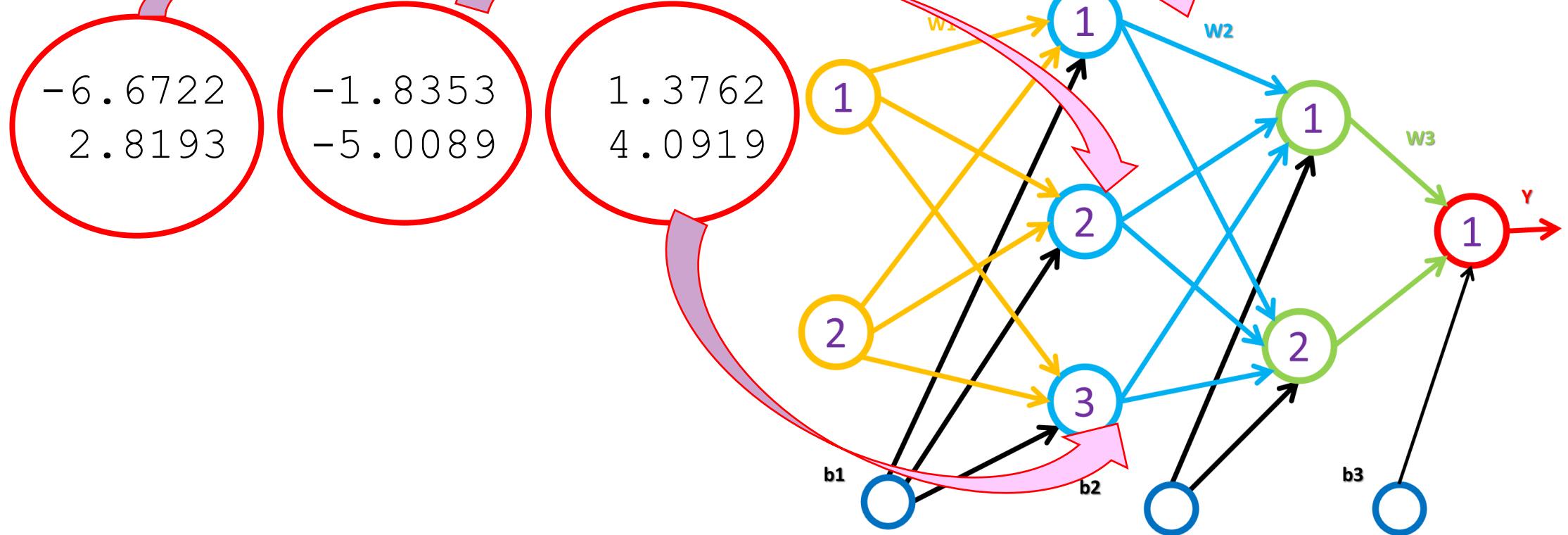
ans =

3×3 logical array



```
>> % Lihat bobot dari H1 ke H2
>> net.LW{2,1}
```

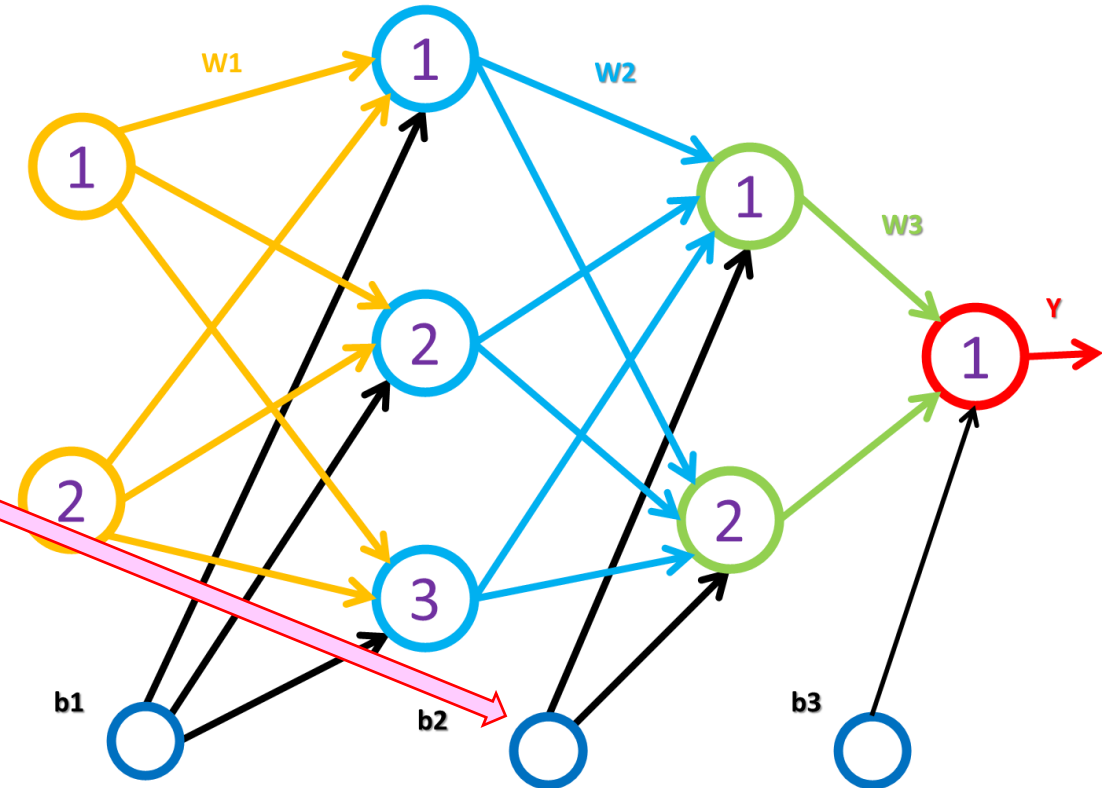
ans =

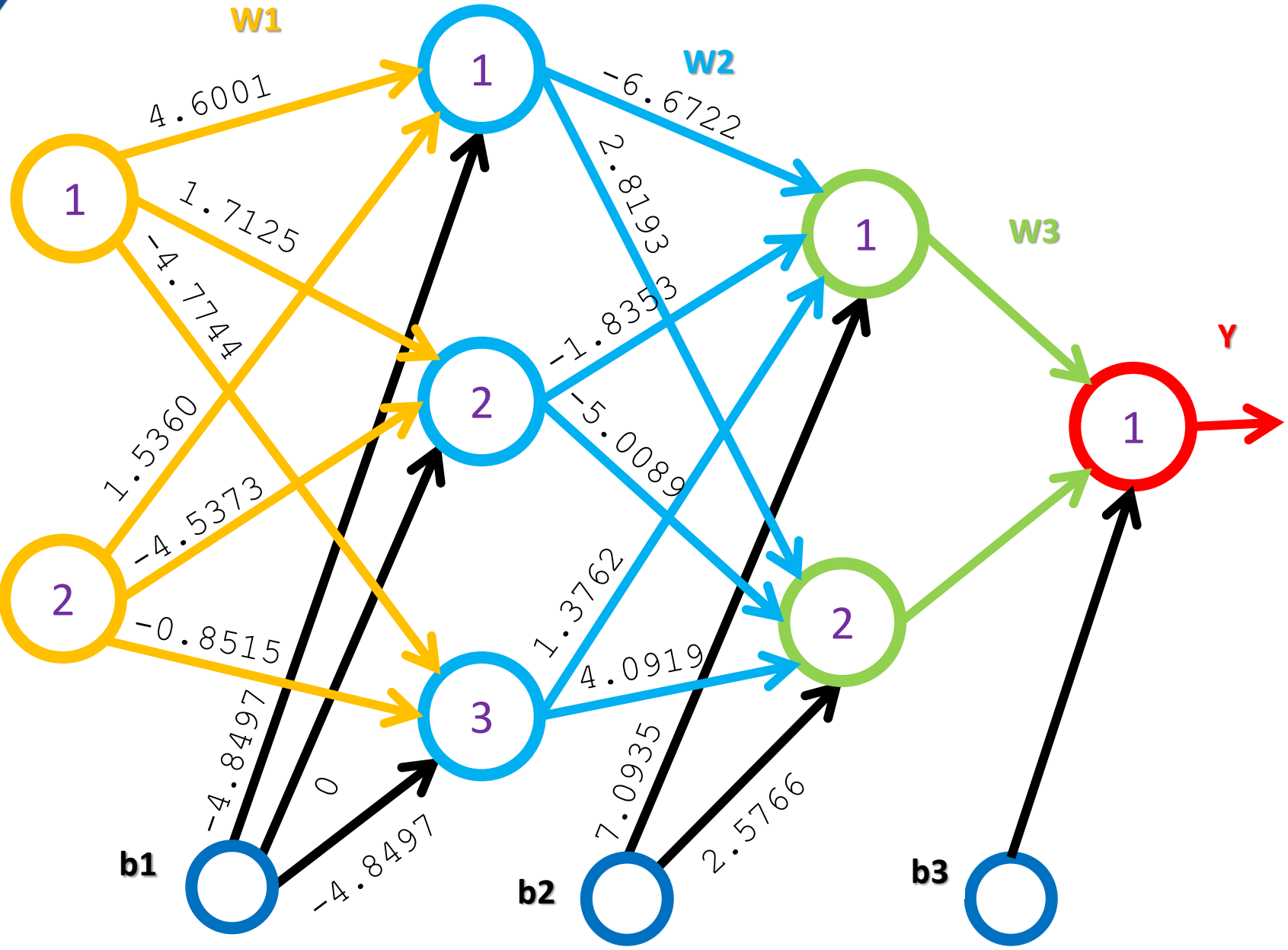


```
>> % Lihat bias di H2  
>> net.b{2}
```

ans =

7.0935
2.5766

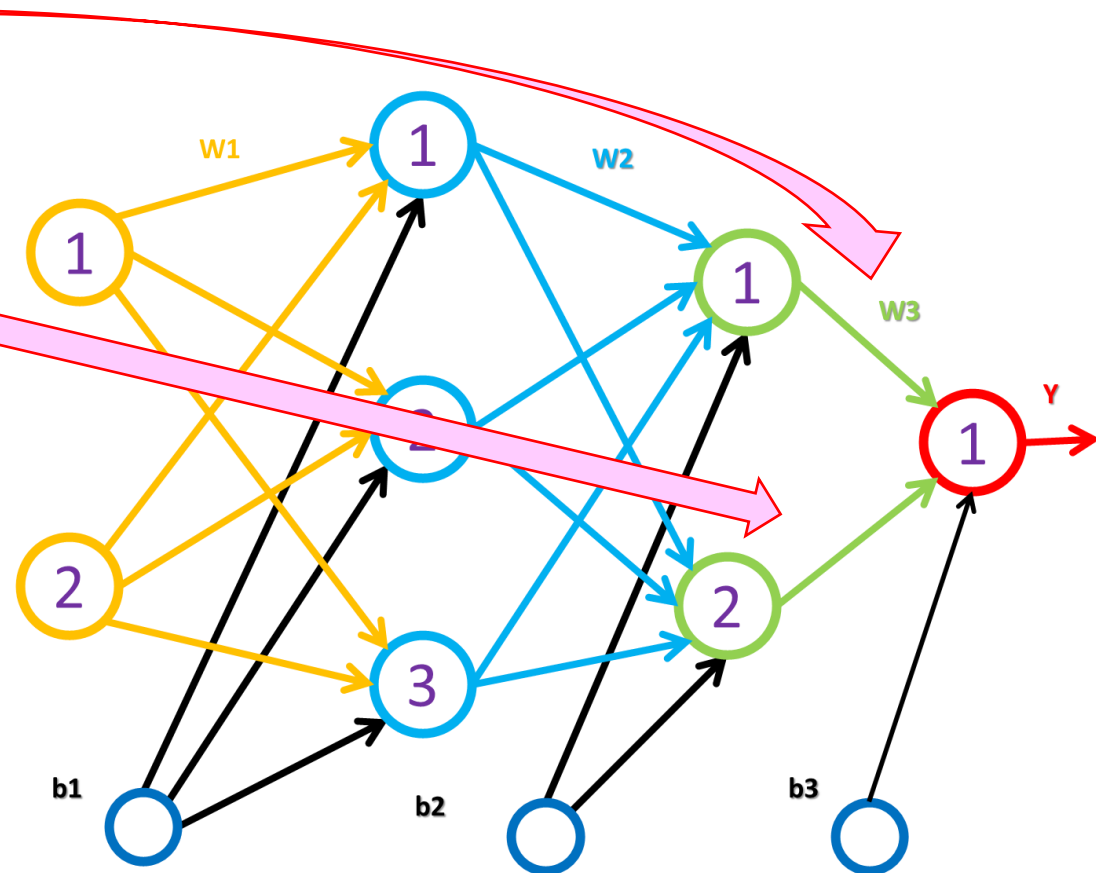




```
>> % Lihat bobot dari H2 ke output  
>> net.LW{3,2}
```

ans =

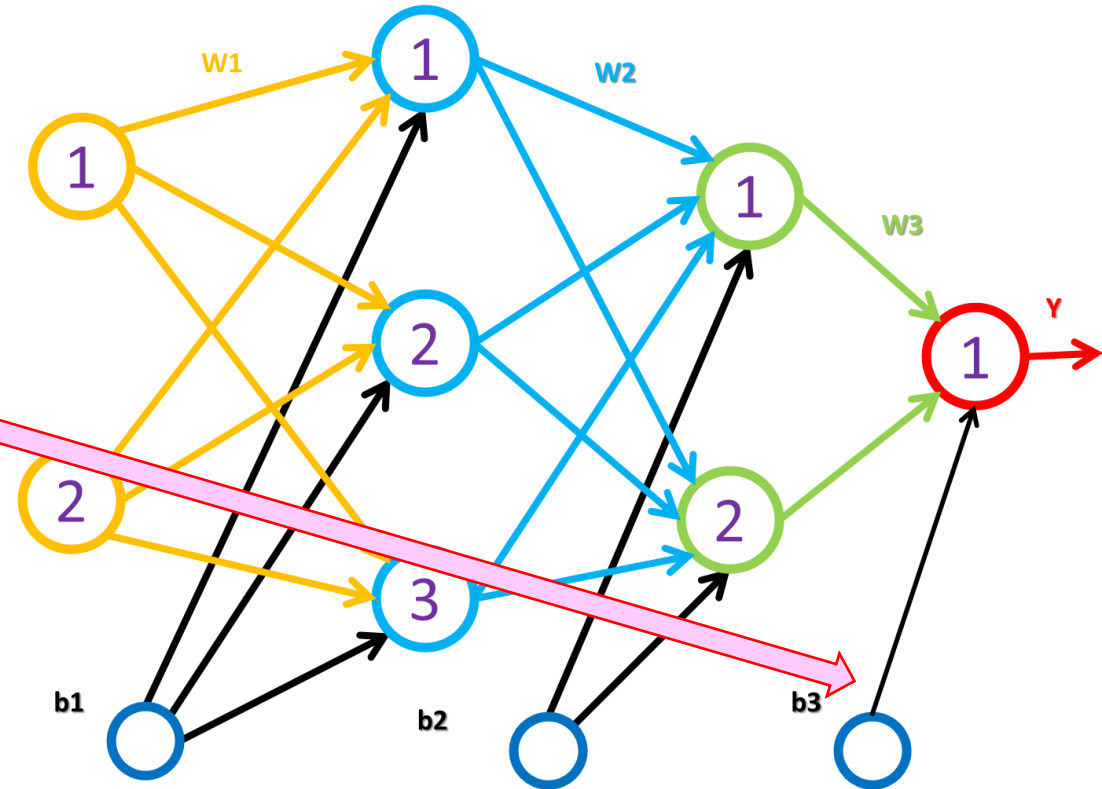
-3.2004 -4.5954

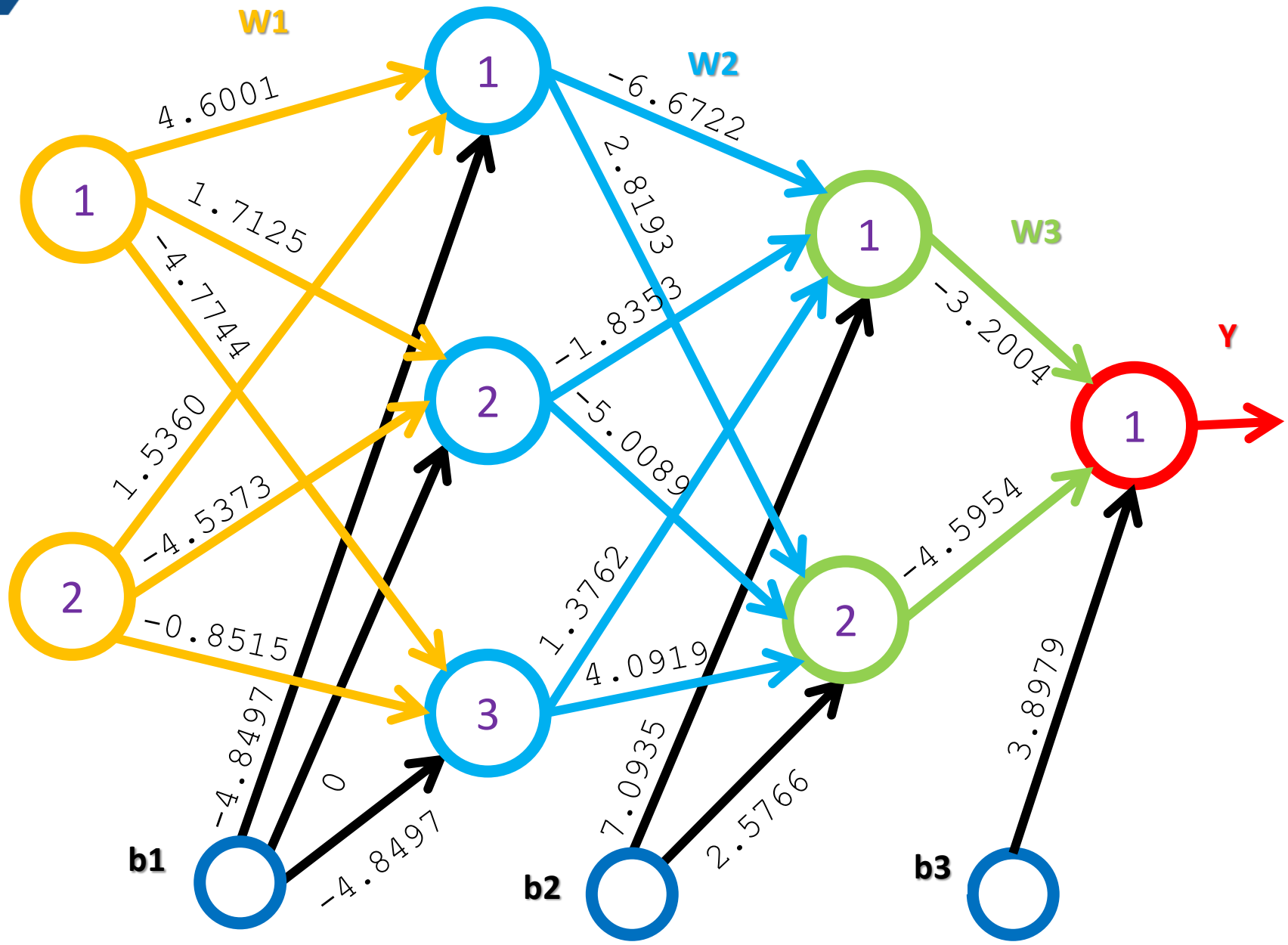


```
>> % Lihat bias di output  
>> net.b{3}
```

ans =

3.8979

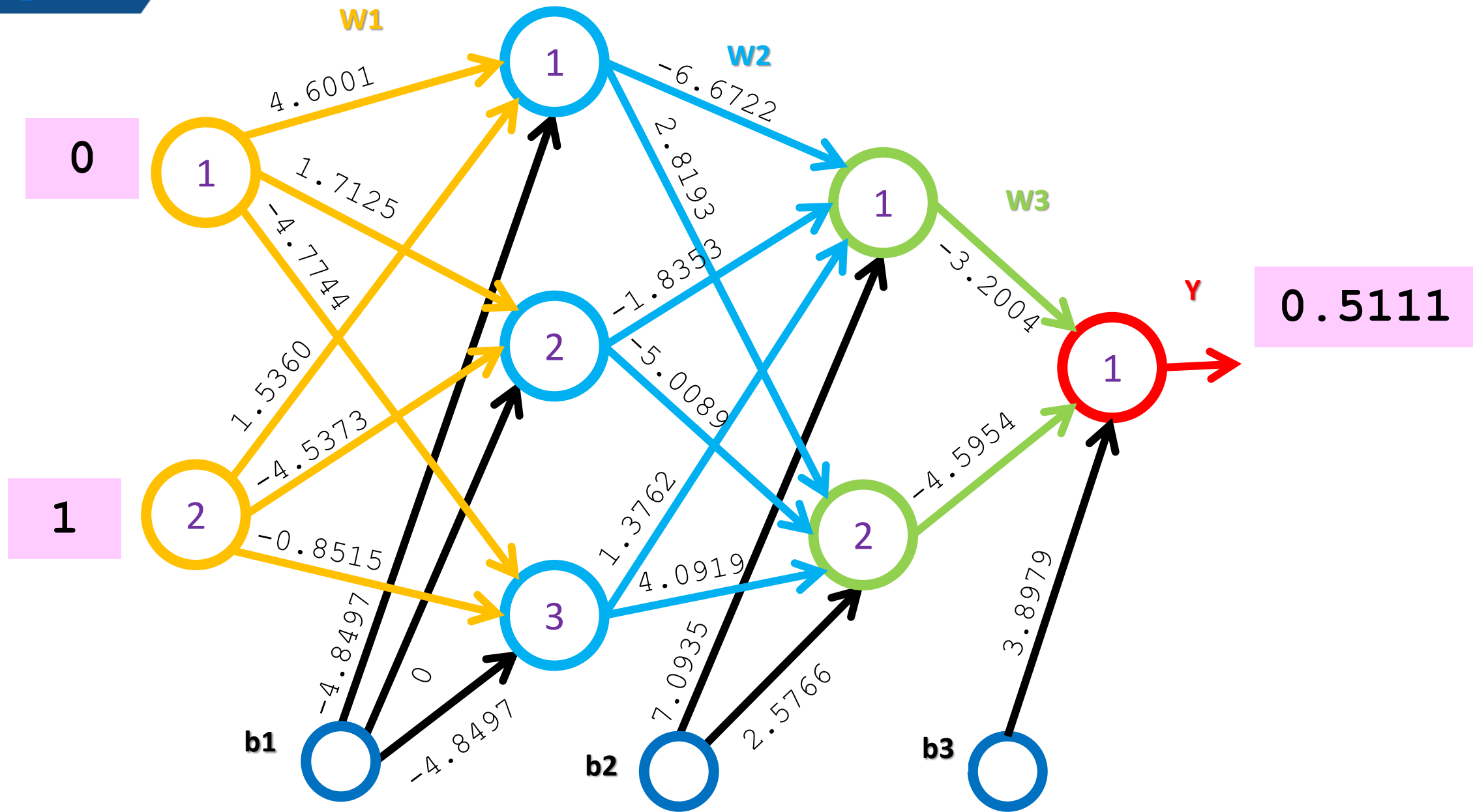


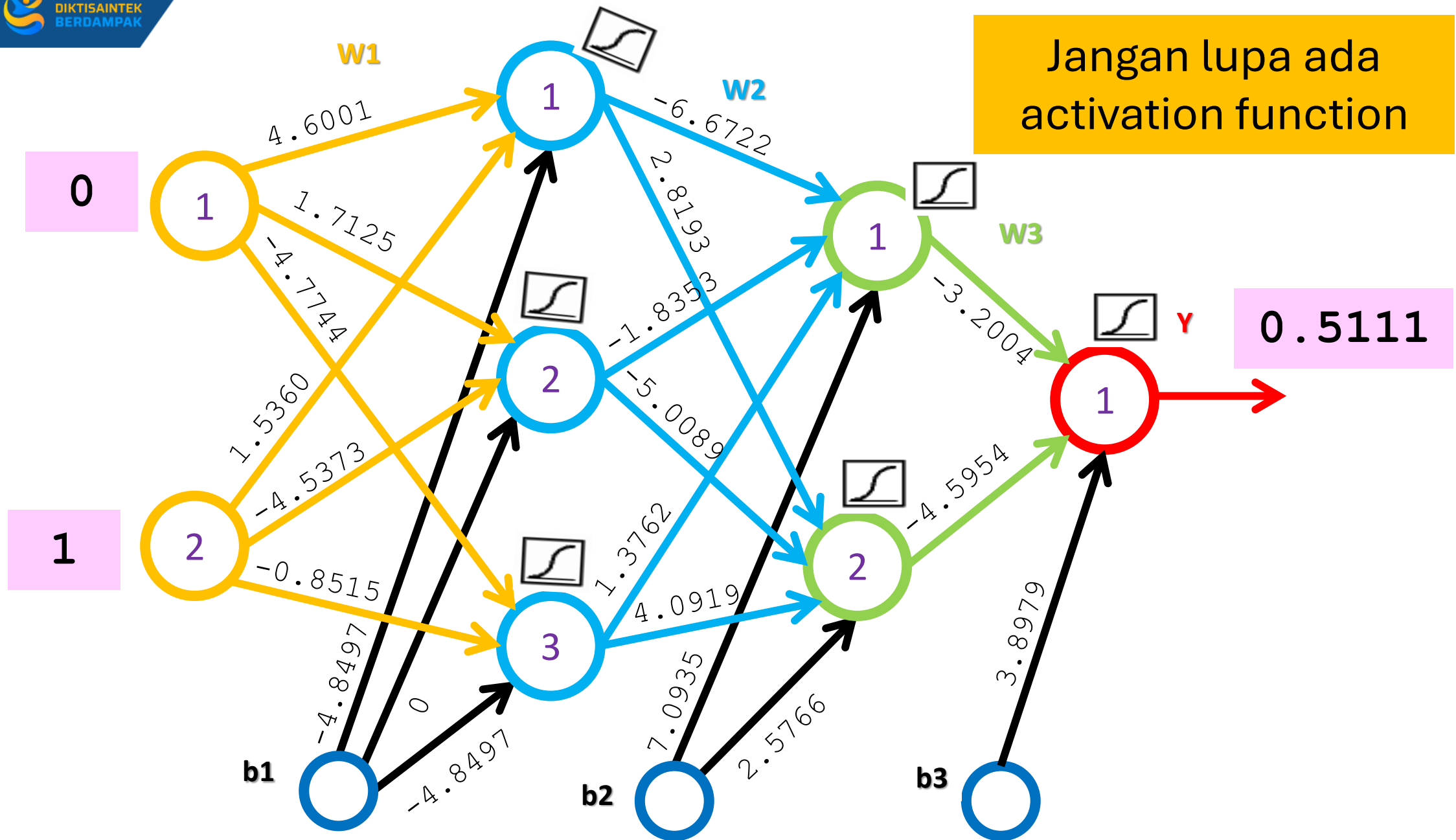



```
>> % Lihat output (sebelum training)  
>> Y = net([0;1])
```

Y =

0.5111





```
% TRAINING
```

```
net = train(net, X, T);
```

```
% EVALUASI
```

```
Y = net(X)
```

```
Y =
```

```
0.5000    0.9997    1.0000    1.0000
```

Tugas

□ Lihat perbaikan bobot dan bias hasil training!